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A quadratic trigonometric weighted spline with local support basis functions



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Abstract A proficient freeform curve design technique is developed by constructing B-spline like local support basis functions using a quadratic trigonometric function introduced. The built basis functions fulfill the properties of the local support, partition of unity and the positivity. The freeform designed curve method, by these constructed local support basis functions, owns the geometric properties such as convex hull and variation diminishing. In the suggested scheme, the Non-uniform freeform curve depiction is presented for deliberation by a transformed Bézier form. The demonstration of the scheme is supportive for various shape effects such as interval tension. As a privilege, the proposed method is C^2 for its default case and is C^1 in general. The proposed method provides a quadratic alternative to the cubic B-splines.

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1. Introduction

Modeling curves, specifically robust curves, which are controllable, well behaved and easily worked out, contribute a special role in geometric modeling and computer graphics. A number of implementations of these vigorous curves in modeling objects, CAD/CAM, font designing, medical imaging, object recognition and fingerprints recognition are the motivations toward curve modeling. Besides the applications, the significance of curve modeling in computer vision, robotics and even

in media is the inspiration of the research. In the existing literature, a wide-ranging effort has been put in the course of curve modeling. For brevity, the readers are referred to [1–27].

In the field of geometric modeling and computer graphics, a cubic spline curve is considered as a decent approach. Although, due to shape control limits, it is not the best choice for designers. Thus one needs to suggest a freeform curve method, which not only provides interested features of shape design, but also keeps decent attributes of cubic spline.

The use of B-splines in geometric designing and modeling is not only a suitable choice and also B-splines are significant in mathematical procedures including curves or surfaces. Also B-splines are the best suitable choice for complex geometry due to their piecewise representation. It also fulfills the properties of non-negativity, partition of unity, variation diminishing and convex hull. Thus it is beneficial to suggest a scheme for a group of quadratic trigonometric splines as a substitute to

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B-splines along with the shape control parameters involved in their depiction. In the beginning, GC^2 splines with local basis were presented by Lewis [10]. In 1981, Barsky [3] constructed Beta splines with basis like B-splines. The geometrical smoothness of the curves modeling was well-maintained by these constructed splines while enabling the continuity conditions at the different knots with various shape control parameters for more flexibility. Later in 1984, Bartels and Beatty [2] built Beta splines local basis that were identical to Boehm's [3] Gamma splines. Foley [7], constructed B-spline like local basis for weighted splines. Nu splines [11,12] were discovered with a cubic spline as a simple substitute of the exponential spline with tension [4,13,21]. In 2004, Sarfraz [22], built B-spline like local support basis for weighted Nu splines. These splines not only recover the weighted splines [7] but also the Nu splines [11,12]. Later in 2008, Sarfraz [23] developed a spline and its free form for designing objects, which recovers Ball's cubic spline. It also owns good characteristics of object designing.

In this paper, a proficient scheme is suggested to build weighted spline like B-spline with a quadratic trigonometric function. The scheme is saturated with meticulous shape effects of parameters. The built local basis has the property of local support and positivity. The curve models constructed with this freeform basis satisfy the geometric properties such as variation diminishing, convex hull, partition of unity and affine invariance. The demonstration of this developed scheme is supportive for a various shape influences such as interval tension. The proposed method is C^2 for its default case and is C^1 in general. The proposed method provides a quadratic alternative to the cubic B-splines.

In Section 2, a quadratic trigonometric function in the weighted form is used to a B-spline curve scheme. As a result of constructing trigonometric B-spline, the free form basis for this trigonometric spline is constructed and proposed in Section 3. In Sections 4, the B-spline like curve scheme is suggested for deliberation by a transformed Bézier form. Furthermore in Section 5, proofs of *variation diminishing*, *convex hull* and *affine invariance* are made, from Bézier form, for freeform quadratic trigonometric weighted curve representation. The demonstration of proposed curve representation in Sections 6 and 7, is supportive for the shape effects like interval tension. In addition shape effects of parameters are well behaved, controllable and display the proposed scheme as a delicate substitute to cubic B-spline.

2. Quadratic trigonometric weighted spline

Let $F_i \in \mathbb{R}^n$ be the values at the knots $t_i, i = 0, 1, \dots, n-1$, which are strictly increasing $t_0 < t_1 < \dots < t_n$. Also $V_i, W_i \in \mathbb{R}^n$. The quadratic trigonometric function for $t \in [t_i, t_{i+1}]$, $i = 0, 1, \dots, n-1$, is defined by

$$P_i(t) = (1 - \sin\theta)^2 F_i + 2(1 - \sin\theta)\sin\theta V_i + 2(1 - \cos\theta)\cos\theta W_i + (1 - \cos\theta)^2 F_{i+1}, \quad (1)$$

where

$$0 \leq \theta \leq \frac{\pi}{2}, \quad \theta(t) = \left(\frac{t - t_i}{h_i} \right) \frac{\pi}{2}, \quad h_i = t_{i+1} - t_i,$$

$$i = 0, 1, \dots, n-1.$$

and

$$V_i = F_i + \frac{h_i}{\pi} M_i, \quad W_i = F_{i+1} - \frac{h_i}{\pi} M_{i+1}. \quad (2)$$

Now rewrite quadratic trigonometric as follows:

$$P_i(t) = A_0(\theta)F_i + A_1(\theta)V_i + A_2(\theta)W_i + A_3(\theta)F_{i+1}, \quad (3)$$

where

$$A_0(\theta) = (1 - \sin\theta)^2, \quad A_1(\theta) = 2(1 - \sin\theta)\sin\theta, \\ A_2(\theta) = 2(1 - \cos\theta)\cos\theta, \quad A_3(\theta) = (1 - \cos\theta)^2,$$

are basis functions with

$$\sum_{j=0}^3 A_j(\theta) = 1. \quad (4)$$

Moreover, these functions are the Bézier functions. The piecewise quadratic trigonometric (1) retains the interpolation properties as follows:

$$P(t_i) = F_i, \quad P(t_{i+1}) = F_{i+1}, \quad P'(t_i) = M_i, \quad P'(t_{i+1}) = M_{i+1}. \quad (5)$$

Then eventually the interpolation of piecewise quadratic trigonometric acts like a Hermite function. Now, by merging the weighted spline characteristics into the quadratic trigonometric spline, the following C^1 constraint equations are acquired:

$$P_i''(t_i) = \frac{w_{i-1}}{w_i} P_{i-1}''(t_i). \quad (6)$$

In this manner, second order continuity at the joints of the curve segments is attained by Eq. (5). In this way, a weighted interpolatory spline is designed. Now the above constraint equations can be represented in the matrix form,

$$\begin{bmatrix} P(t_{i+}) \\ P'(t_{i+}) \\ P''(t_{i+}) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{w_{i-1}}{w_i} \end{bmatrix} \begin{bmatrix} P(t_{i-}) \\ P'(t_{i-}) \\ P''(t_{i-}) \end{bmatrix}, \quad (7)$$

which results as a system of consistency equations in M_i 's (unknowns). By using (2) and (6) we have

$$\frac{\pi w_{i-1}}{2w_i h_{i-1}} M_{i-1} + \left(\frac{\pi}{h_i} + \frac{\pi w_{i-1}}{h_{i-1} w_i} \right) M_i + \frac{\pi}{2h_i} M_{i+1} \\ = \frac{\pi^2}{2h_i} \Delta_i + \frac{\pi^2 w_{i-1}}{2h_{i-1} w_i} \Delta_{i-1}, \quad (8)$$

where

$$\Delta_i = \frac{F_{i+1} - F_i}{h_i}.$$

For suitable end conditions with the constraint on the shape parameter $w_i = 1, \forall i$, the tri-diagonal linear system in (8) for derivatives M_i 's, is diagonally dominant.

3. Formation of free form basis for a trigonometric weighted spline

Consider a local function $\xi_j(t)$ to build basis functions like B-spline for the proposed quadratic trigonometric spline. For this purpose consider additional knots $t_{-3} < t_{-2} < t_{-1} < t_0$ and $t_n < t_{n+1} < t_{n+2} < t_{n+3}$ on both sides of the interval $[t_0, t_n]$. The local function $\xi_j(t), j = -1, \dots, n+2$ is defined by

$$\xi_j(t) = \begin{cases} 0 & t < t_{j-2}, \\ 1 & t \geq t_{j+1}. \end{cases} \quad (9)$$

whereas on the left over three intervals $[t_i, t_{i+1}]$, $i = j - 2, j - 1, j$, $\xi(t)$ has the quadratic trigonometric form defined by (1):

$$\xi_j(t) = A_0(\theta)\widehat{F}_{j,i} + A_1(\theta)\widehat{V}_{j,i} + A_2(\theta)\widehat{W}_{j,i} + A_3(\theta)\widehat{F}_{j,i+1}. \quad (10)$$

Here $A_k(\theta)$, $k = 0, \dots, 3$, are defined as in (3). Also

$$\begin{aligned} \widehat{F}_{j,i} &= \xi_j(t_i), \quad \widehat{V}_{j,i} = \xi_j(t_i) + \frac{h_i}{\pi} \xi'_j(t_i), \\ \widehat{W}_{j,i} &= \xi_j(t_{i+1}) - \frac{h_i}{\pi} \xi'_j(t_{i+1}) \end{aligned} \quad (11)$$

Now, by applying the constraint $w_i = 1, \forall i$ on $\xi_j(t)$, we have obtained the following:

$$\left. \begin{aligned} \xi_j(t_{j-2}) &= 0, \xi'_j(t_{j-2}) = 0, \xi''_j(t_{j-2}) = 0, \\ \xi_j(t_{j-1}) &= \rho_{j-1}, \xi'_j(t_{j-1}) = \hat{\rho}_{j-1}, \\ \xi_j(t_j) &= 1 - \tau_j, \xi'_j(t_j) = \hat{\tau}_j, \end{aligned} \right\} \quad (12)$$

where

$$\begin{aligned} \tau_j &= \frac{h_j}{\pi} \hat{\tau}_j, \\ \rho_{j-1} &= \frac{h_{j-2}}{\pi} \hat{\rho}_{j-1}, \\ \hat{\tau}_j &= \frac{u_3 v_1 - u_1 v_3}{u_2 v_1 - u_1 v_2}, \\ \hat{\rho}_{j-1} &= \frac{u_3}{u_1} - \frac{u_2}{u_1} \hat{\tau}_j, \\ u_1 &= \frac{1}{w_{i-1}} + \frac{\pi}{h_{i-1}} + \frac{\pi}{2h_{i-2}} \frac{w_{i-2}}{w_{i-1}} + \frac{\pi}{2h_{i-1}^2} h_{i-2}, \\ u_2 &= \frac{\pi}{2h_{i-1}} + \frac{\pi}{2h_{i-1}^2} h_i, \\ u_3 &= \frac{\pi^2}{2h_{i-1}^2}, \\ v_1 &= \frac{\pi}{2h_{i-1}} \frac{w_{i-1}}{w_i} + \frac{\pi}{2h_{i-1}^2} \frac{w_{i-1}}{w_i} h_{i-2}, \\ v_2 &= \frac{1}{w_i} + \frac{\pi}{2h_i} + \frac{\pi}{h_{i-1}} \frac{w_{i-1}}{w_i} + \frac{\pi}{2h_{i-1}^2} \frac{w_{i-1}}{w_i} h_i, \\ v_3 &= \frac{\pi^2}{2h_{i-1}^2} \frac{w_{i-1}}{w_i}. \end{aligned}$$

Now, from the difference of $\xi_j(t)$ functions, the free form weighted spline basis is obtained as

$$N_j(t) = \xi_j(t) - \xi_{j+1}(t), \quad j = -1, \dots, n+1. \quad (13)$$

Therefore for every $[t_i, t_{i+1}]$ if

$$N_j(t) = A_0(\theta)F_{j,i} + A_1(\theta)V_{j,i} + A_2(\theta)W_{j,i} + A_3(\theta)F_{j,i+1}, \quad (14)$$

where

$$\begin{aligned} F_{j,i} &= N_j(t_i), \quad V_{j,i} = N_j(t_i) + \frac{h_i}{2} N'_j(t_i), \\ W_{j,i} &= N_j(t_{i+1}) - \frac{h_i}{2} N'_j(t_{i+1}), \end{aligned}$$

then

$$\left. \begin{aligned} N_j(t_i) &= 0, N'_j(t_i) = 0, & \text{for } i \neq j-1, j, j+1, \\ N_j(t_{j-1}) &= \rho_{j-1}, & N'_j(t_{j-1}) = \hat{\rho}_{j-1}, \\ N_j(t_j) &= 1 - \tau_j - \rho_j, & N'_j(t_j) = \hat{\tau}_j - \hat{\rho}_j, \\ N_j(t_{j+1}) &= \tau_{j+1}, & N'_j(t_{j+1}) = -\hat{\tau}_{j+1}. \end{aligned} \right\} \quad (15)$$

Proposition 1. The free form quadratic trigonometric weighted spline (QTWS) basis functions have the geometric properties as follows:

• **Local support property:**

$$N_j(t) = 0, \quad \text{for } t \notin (t_{j-2}, t_{j+1}), \quad (16)$$

• **Partition of unity property:**

$$\sum_{j=-1}^{n+1} N_j(t) = 1, \quad t \in [t_0, t_n]. \quad (17)$$

• **Property of Positivity:**

$$N_j(t) \geq 0, \quad \forall t. \quad (18)$$

Proof. From the definition and construction of the basis functions, the above properties follow straightaway. $\square$

4. B-spline like curve modeling

For freeform curve representation, it is requisite to compute the following

$$P(t) = \sum_{j=-1}^{n+1} d_j N_j(t) \quad \text{for } t \in [t_0, t_n], \quad (19)$$

with the control points $d_j \in \mathbb{R}^n$. Now by Proposition 1, rewrite Eq. (18) as follows:

$$P(t) = \sum_{j=i-1}^{i+2} d_j N_j(t), \quad t \in [t_i, t_{i+1}], \quad i = 0, 1, \dots, n-1. \quad (20)$$

After some extra operations, (20) converts to the piecewise Bézier form attained as follows:

$$P(t) = A_0(\theta)F_i + A_1(\theta)V_i + A_2(\theta)W_i + A_3(\theta)F_{i+1}, \quad (21)$$

where

$$\left. \begin{aligned} F_i &= \tau_i d_{i-1} + (1 - \tau_i - \rho_i) d_i + \rho_i d_{i+1}, \\ V_i &= \psi_i d_{i-1} + (1 - \psi_i - \phi_i) d_i + \phi_i d_{i+1}, \\ W_i &= \phi_i d_i + (1 - \phi_i - \Psi_i) d_{i+1} + \Psi_i d_{i+2}. \end{aligned} \right\} \quad (22)$$

with

$$\begin{aligned} \psi_i &= \tau_i - \hat{\tau}_i \frac{h_i}{\pi}, \quad \phi_i = \rho_i + \hat{\rho}_i \frac{h_i}{\pi} \quad \text{and} \quad \phi_i \\ &= \tau_{i+1} + \hat{\tau}_{i+1} \frac{h_i}{\pi}, \quad \Psi_i = \rho_{i+1} - \hat{\rho}_{i+1} \frac{h_i}{\pi}. \end{aligned} \quad (23)$$

Let $P_i = [F_i V_i W_i F_{i+1}]^T$, $R_i = [d_{i-1} d_i d_{i+1} d_{i+2}]^T$ and

$$Q_i = \begin{bmatrix} \tau_i & 1 - \tau_i - \rho_i & \rho_i & \\ \psi_i & 1 - \phi_i - \psi_i & \phi_i & \\ & \phi_i & 1 - \phi_i - \Psi_i & \Psi_i \\ & \tau_{i+1} & 1 - \tau_{i+1} - \rho_{i+1} & \rho_{i+1} \end{bmatrix}.$$

The matrix construction of (22) is as follows:

$$P_i = Q_i R_i. \quad (24)$$

Now, the transformed Bézier in (24) can easily be used for computational purpose.

Remark 1. It is noteworthy that the interpolation problem of freeform QTWS can be sorted by

$$\sum_{j=-1}^{n+1} d_j N_j(t_i) = F_i \forall i, \quad (25)$$

with $N_j(t_i)$, a tridiagonal matrix, diagonally dominant as $0 < \rho_i, \tau_i < \frac{1}{2}$. Thus a unique QTWS exists as a substitute to B-spline.

5. Geometric properties of free form QTWS

Proposition 2 (Convex Hull Property). The freeform QTWS curve entirely lies within the convex hull determined by its control polygon $\{F_i V_i W_i F_{i+1}\}$.

Proof. Since from (17) and (18),

$$\sum_{j=-1}^{n+1} N_j(t) = 1 \quad \text{for } t \in [t_0, t_n]$$

and

$$N_j(t) \geq 0, \quad \forall t,$$

thus the freeform QTWS curve entirely lies within the convex hull determined by its control polygon $\{F_i V_i W_i F_{i+1}\}$ as displayed in Fig. 1(a) and (b). $\square$

Remark 1. The convex hull property disrupts for negative values of the parameters w_k 's, where k is any interval.

Fig. 2(a), (b) and (c) demonstrates the violation of convex hull property for $w_k = -1.7$ in the 10th interval, $w_k = -1.2$ in the 7th interval and $w_k = -1.4$ in the 3rd interval respectively.

Proposition 3 (Variation Diminishing Property). Let $P(t) = \sum_{j=0}^3 d_j N_j(t)$ be freeform QTWS curve for $t \in [t_i, t_{i+1}]$. Any hyper plane with dimension $N - 1$ will cross the freeform QTWS curve no more times it will cross the control polygon determined by the control points $d_j = \{F_i, V_i, W_i, F_{i+1}\} \in \mathbb{R}^n$.

Proof. From (23) it can be revealed that $\phi_i \geq 0, \phi_i \geq 0$ and $\phi_i + \phi_i \leq 1$. Also it can be noted from (22) that W_i and V_i lie on the line linking d_i and d_{i+1} and W_i is after V_i . Since

$$\delta_i = \frac{h_{i-1}}{h_i + h_{i-1}} \quad (26)$$

holds and $0 < \delta_i < 1$. Then the following can be written as follows:

$$F_i = (1 - \delta_i) W_{i-1} + \delta_i V_i. \quad (27)$$

Hence, the control polygon of piecewise defined Bézier is attained through the corner cutting of freeform QTWS control polygon. Since the piecewise Bézier satisfies the variation diminishing property, thus, the freeform QTWS fulfills variation diminishing property.

Fig. 3(a) and (b) illustrates the variation diminishing property. $\square$

Remark 2. The variation diminishing property violates, for $w_k < 0$, for any interval k , as revealed in Fig. 4(a), (b) and (c).

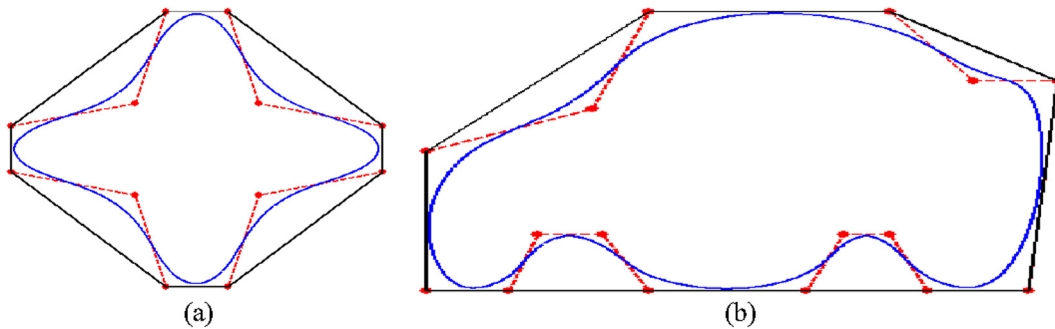


Fig. 1 Convex Hull property.

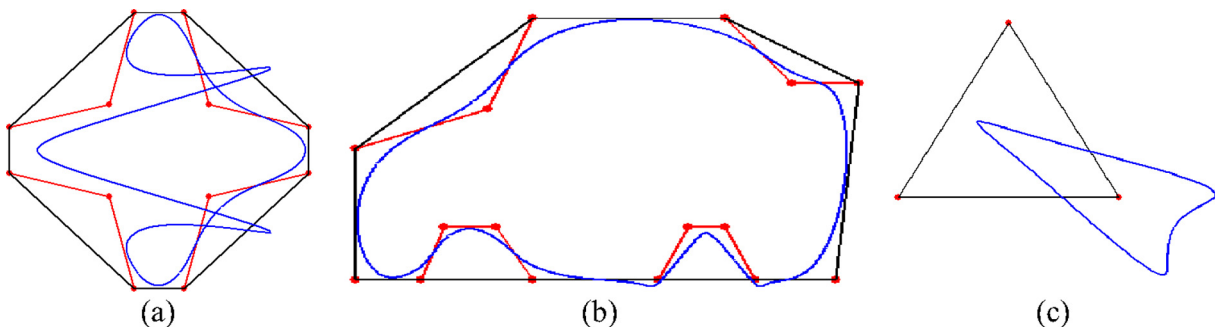


Fig. 2 Violation of Convex Hull property.

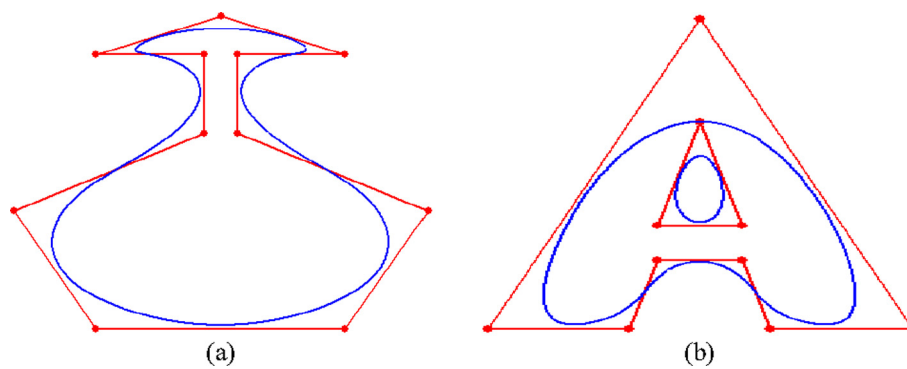


Fig. 3 Variation diminishing property.

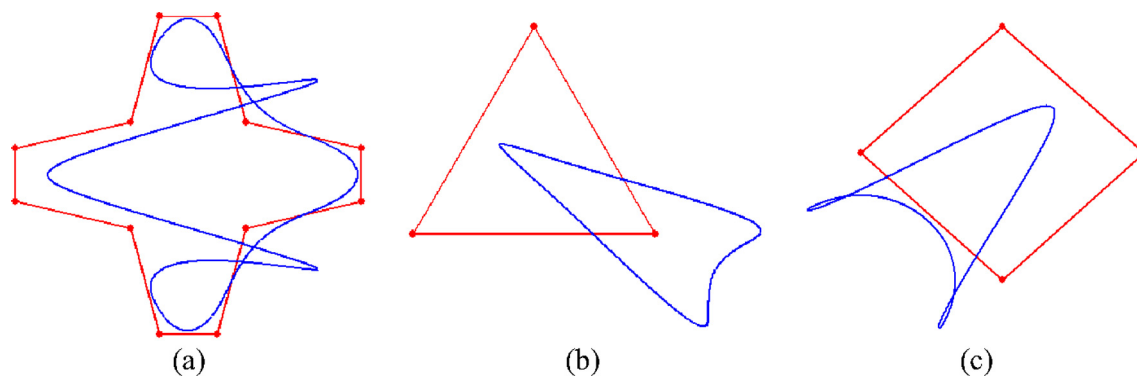


Fig. 4 Violation of variation diminishing property.

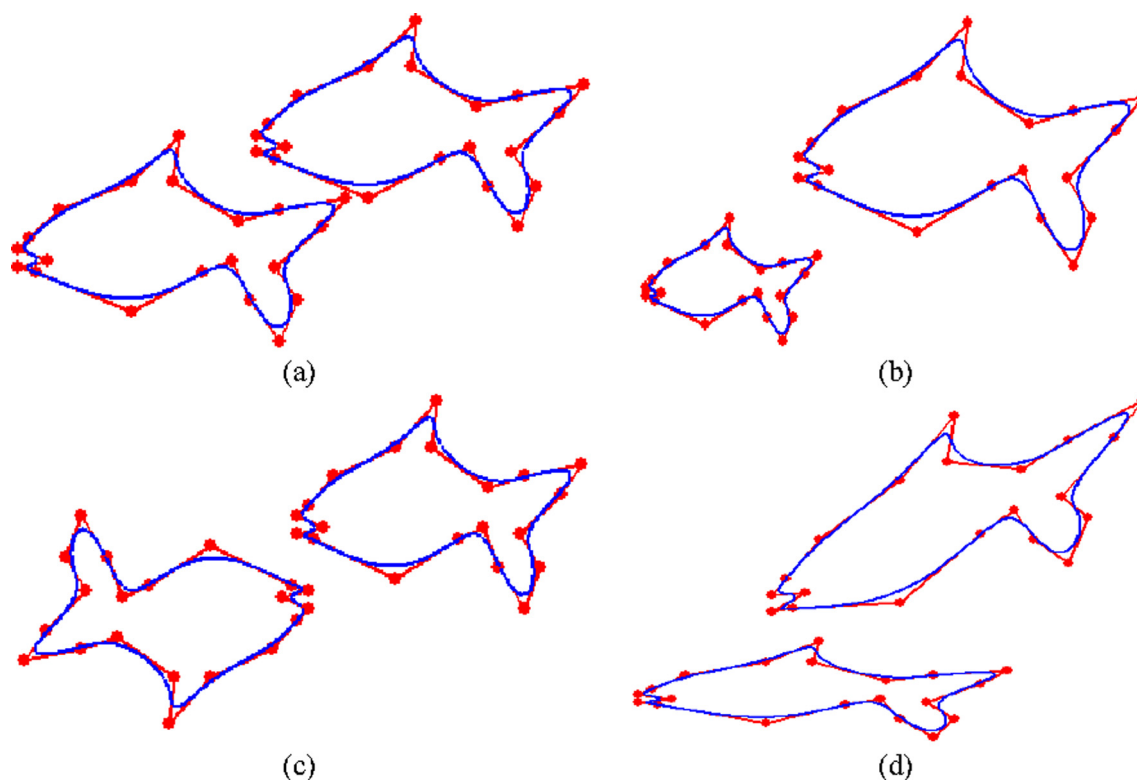


Fig. 5 Affine invariant under (a) translation, (b) scaling, (c) rotation, (d) shearing.

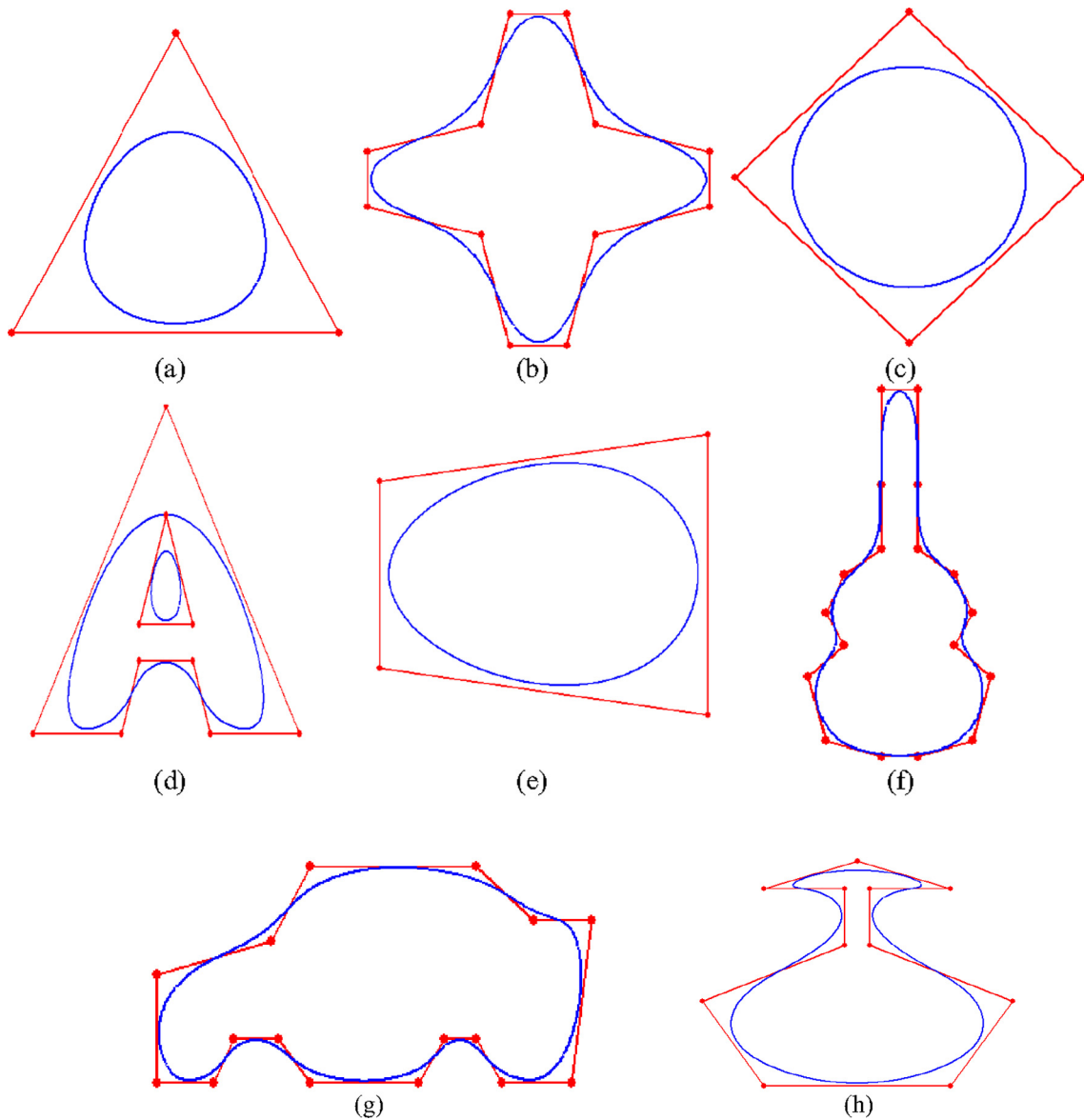


Fig. 6 The data points of various existing objects (a), (b), (c), (d), (e), (f), (g) and (h), are taken and then interpolated by freeform QTWS using periodic end conditions.

Fig. 4(a), (b) and (c) violates variation diminishing property for $w_k = -1.7$ in the 10th interval, $w_k = -1.4$ in the 3rd interval and $w_k = -1.4$ in the 4th interval, respectively.

Proposition 4 (*Affine Invariance Property*). The freeform QTWS curve $P(t) = \sum_{j=0}^3 d_j N_j(t)$ for $t \in [t_i, t_{i+1}]$ with the control points $d_j = \{F_i, V_i, W_i, F_{i+1}\} \in \mathbb{R}^n$ is affine invariant.

Proof. Consider an affine transformation T , given by

$$T(x, y) = (ax + by + c, dx + ey + f),$$

and let the freeform QTWS curve $P(t) = \sum_{j=0}^3 d_j N_j(t)$ for $t \in [t_i, t_{i+1}]$ has control points $d_j(p_j, q_j), j = 0, \dots, 3$. Then

$$P(t) = (x(t), y(t)) = \left(\sum_{j=0}^3 p_j N_j(t), \sum_{j=0}^3 q_j N_j(t) \right).$$

Now,

$$T(P(t)) = \left(a \sum_{j=0}^3 p_j N_j(t) + b \sum_{j=0}^3 q_j N_j(t) + c, d \sum_{j=0}^3 p_j N_j(t) + e \sum_{j=0}^3 q_j N_j(t) + f \right).$$

Since, from partition of unity, we have $\sum_{j=0}^3 N_j(t) = 1$ for $t \in [t_i, t_{i+1}]$. Thus

$$\begin{aligned} T(P(t)) &= \sum_{j=0}^3 (ap_j + bq_j + c) N_j(t), \sum_{j=0}^3 (dp_j + eq_j + f) N_j(t) \\ &= \sum_{j=0}^3 (ap_j + bq_j + c, dp_j + eq_j + f) N_j(t) \\ &= \sum_{j=0}^3 T(d_j) N_j(t). \end{aligned}$$

Illustration of the above Proposition, is made in Fig. 5(a), (b) and (c). Fig. 5(d) shows shearing transformation effects. $\square$

6. Shape properties

The freeform QTWS is particularly beneficial for those objects which require the changes in the specific portion of the objects. And one can made these changes by varying w_i 's in the particular region using interval tension.

Proposition 5 (*Interval Tension Property*). Let $w_i = 1 \forall i, i \neq k$ and $w_k \geq 1$ on $[t_k, t_{k+1}]$ for any k . Consider

$$L_k = (1 - \rho)d_k + \rho d_{k+1},$$

$$L_{k+1} = \tau d_k + (1 - \tau)d_{k+1}.$$

L_k and L_{k+1} defined the two distinct points lie on the line linking d_k and d_{k+1} of the control polygon. Also,

$$\rho = \lim_{w_k \rightarrow \infty} \rho_k \text{ and } \tau = \lim_{w_k \rightarrow \infty} \tau_{k+1},$$

which is clear from (12) that ρ_k and τ_k are the functions of w_k . Since from (19), $P(t)$ is freeform QTWS then for $w_k \rightarrow \infty$, the freeform QTWS uniformly converges to $L(t)$ on $[t_k, t_{k+1}]$ and

$$L(t) = (1 - \theta)L_k + \theta L_{k+1}, \quad \theta(t) = \frac{t - t_k}{h_k}.$$

Proof. From (12) it can be noted that $\rho + \tau < 1$ and L_k are before L_{k+1} .

As $w_k \rightarrow \infty$, then $\hat{\tau}_{k+1} \rightarrow 0$, thus $\hat{\rho}_k \rightarrow 0$.

Hence

$$\lim_{w_k \rightarrow \infty} \rho_k = 0 = \lim_{w_k \rightarrow \infty} \tau_{k+1}.$$

Now by using (22), it can be observed that

$$\lim_{w_k \rightarrow \infty} F_k = L_k \text{ and } \lim_{w_k \rightarrow \infty} F_{k+1} = L_{k+1}.$$

Moreover, the piecewise Bézier in (21) can also be re-written as follows:

$$P(t) = l_k(t) + e_k(t; r_k), \quad t \in [t_k, t_{k+1}],$$

where

$$l_k(t) = (1 - \theta)F_k + \theta F_{k+1}, \quad \theta(t) = \frac{t - t_k}{h_k}, \quad \lim_{w_k \rightarrow \infty} \|e_k\| = 0.$$

Now, by the global tension property, we have

$$\lim_{w_k \rightarrow \infty} \|L_k - l_k\| \leq \lim_{w_k \rightarrow \infty} \|L - P\| + \lim_{w_k \rightarrow \infty} \|e_k\| = 0 \text{ on } [t_k, t_{k+1}],$$

which concludes the result. $\square$

Remark 3. It is worth mentioning that the changes made by the freeform QTWS, are local but not global. To be precise, freeform QTWS is modest overall.

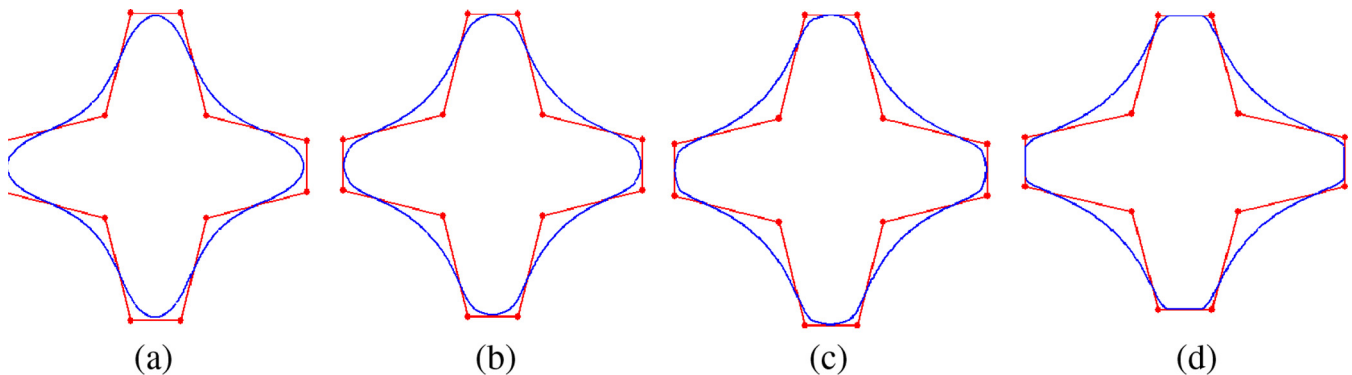


Fig. 7 The interval tension property is observed on four side intervals in (a), (b) and (c) by varying $w_i = 1, 5, 10$ and 100 , respectively, in the freeform QTWS.

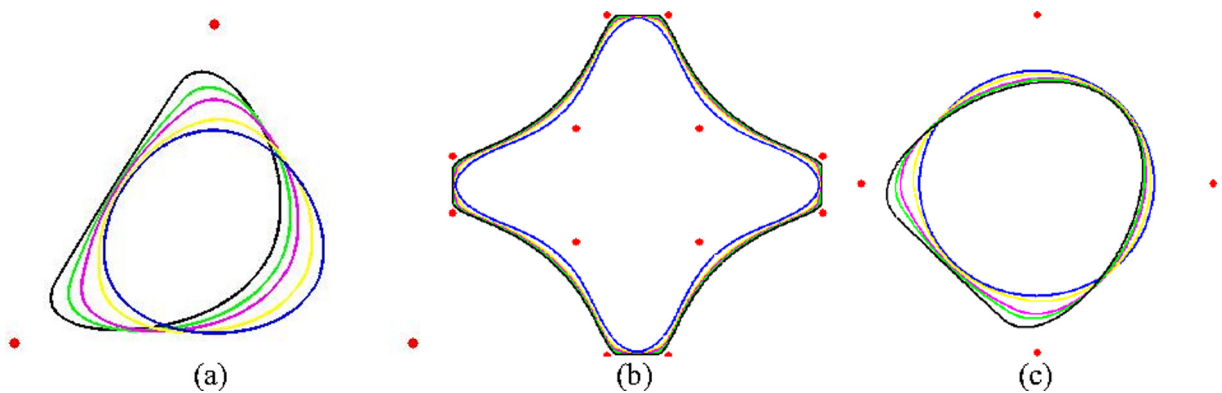


Fig. 8 The interval tension property is displayed in different intervals of the objects (a), (b) and (c) by varying $w_i = 1, 2, 5, 10$ and 100 in the freeform QTWS.

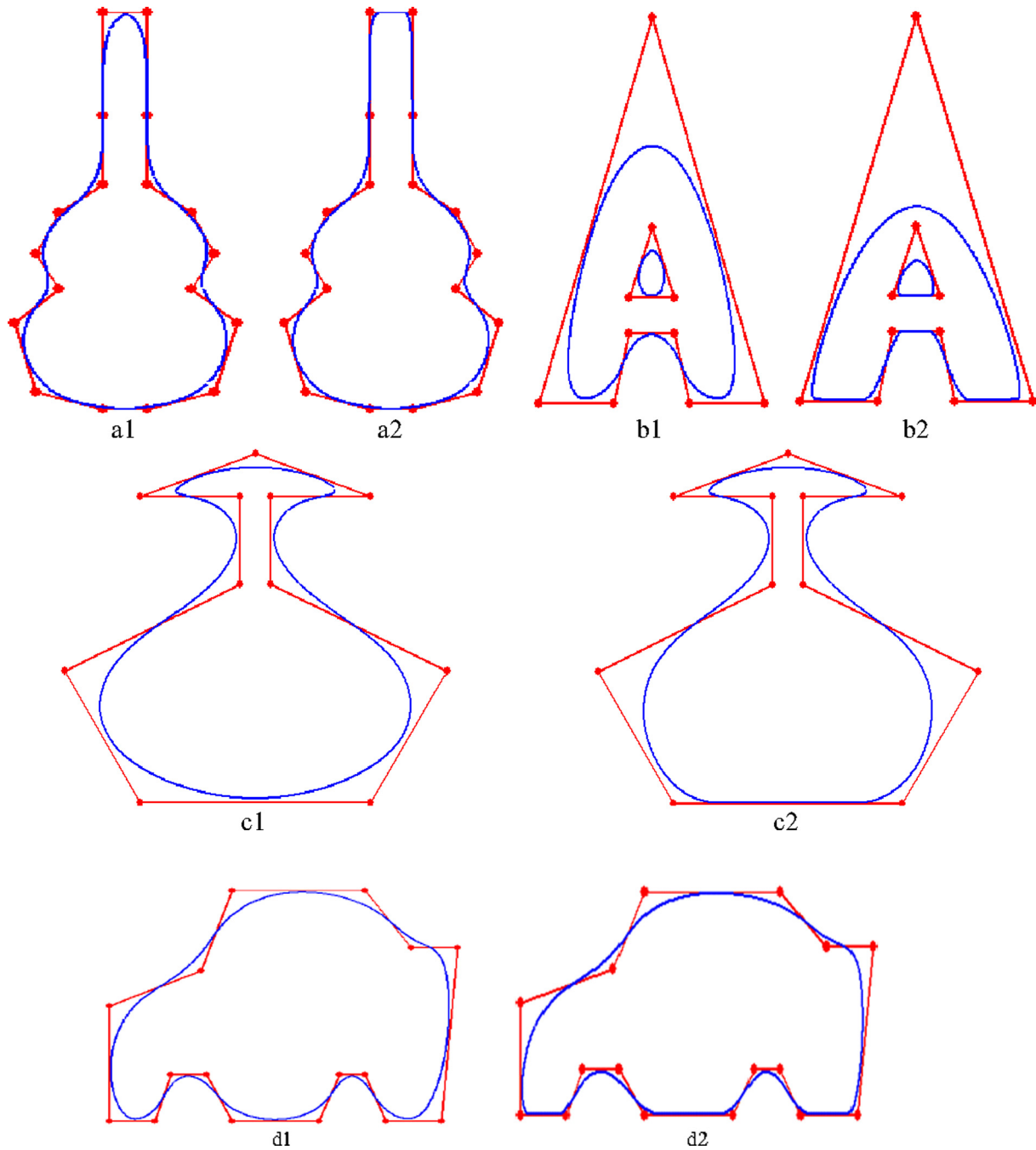


Fig. 9 The curves in a1, b1, c1 and c1 are the default figures using trigonometric weighted B-spline, whereas a2, b2, c2 and c2, respectively, are their modified forms by changing the value(s) of parameter(s) w_i in the intervals wherever changes are demonstrated.

7. Demonstration

For the set of data points $(x, y) \in \mathbb{R}^2$, tension behavior of freeform QTWS is elaborated with various examples. In the demonstration, the default value of parameter $w_i = 1$, is taken, unless it is stated.

In Fig. 6(a-h), 2D data from real life objects are chosen and then interpolated by proposed freeform QTWS using periodic end conditions. It is noted that the shape of the objects is invariant by varying $w_i, \forall i$. To be precise, freeform QTWS is modest globally.

In Fig. 7(a), (b), (c) and (d), the interval tension property is shown on four side intervals (left, right, top and bottom) by varying $w_i = 1, 5, 10$ and 100, respectively, in the freeform QTWS.

In Fig. 8(a), (b) and (c), the interval tension property is presented in different intervals of different shapes by varying $w_i = 1, 2, 5, 10$ and 100 in the freeform QTWS. It reflects a continuous growth of tensions for the same object.

Fig. 9 also demonstrates shape effects for different objects by comparing with their default trigonometric quadratic weighted B-spline curves in Fig. 9(a1), (b1), (c1) and (d1).

The corresponding modified curves are shown in Fig. 9(a2), (b2), (c2) and (d2), respectively using interval tension properties.

8. Advantages and features of constructed scheme

The constructed scheme has various features and benefits as follows:

- It enjoys decent features of trigonometric splines.
- It keeps the best suitable geometric properties of B-splines.
- The proposed scheme accomplishes the C^2 smoothness for its default case and is C^1 in general.
- It is also rich with interested shape designing features.
- In this scheme, the B-spline like curve is presented for deliberation by a transformed Bézier form.
- The proposed scheme recovers the quadratic trigonometric B-spline.
- It has one family of shape parameters, which contributes to various shape effects with interval tension.
- The proposed method provides a quadratic alternative to the cubic B-splines as it possesses four control points, for each piece of curve, in its description.

9. Conclusion

A free form QTWS curve scheme has been constructed with the purpose of curve modeling. This method has not only the ability of curve modeling, but also it is an appropriate approximation to the control points. This scheme has been elaborated with the view of its implementations in geometric modeling, CAGD and computer graphics. Moreover these shapes are adequate and particularly helpful for variety of shape effects, like interval tension. The proposed method is C^2 for its default case and is C^1 in general. The proposed method provides a quadratic alternative to the cubic B-splines.

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References

- [1] R. Bartels, J. Beatty, Beta-splines with a difference Technical Report CS-83-40, Computer Science Department, University of Waterloo, Waterloo, Canada, 1984.
- [2] W. Boehm, Curvature continuous curves and surfaces, *Comput. Aided Geom. Design* 2 (2) (1985) 313–323.
- [3] B. Barsky, The Beta-spline: A Local Representation Based on Shape Parameters and Fundamental Geometric Measure Ph.D. Thesis, University of Utah, 1981.
- [4] A. Cline, Curve fitting in one and two dimensions using splines under tension, *Commun. ACM* 17 (1974) 218–223.
- [5] P. Dierckx, B. Tytgat, Generating the Bézier points of b-spline curve, *Comput. Aided Geomet. Design* 6 (1989) 279–291.
- [6] G.E. Farin, *Curves, surfaces for CAGD*, Academic Press, New York, 1996.
- [7] T.A. Foley, Local control of interval tension using weighted splines, *Comput. Aided Geomet. Design* 3 (1987) 281–294.
- [8] T.A. Foley, Interpolation with interval and point tension controls using cubic weighted v -splines, *ACM Trans. Math. Software* 13 (1987) 68–96.
- [9] T.N.T. Goodman, K. Unsworth, Generation of Beta spline curves using a recursive relation, *Fundam. Algorithms Comput. Graph.* (1985) 326–357.
- [10] J. Lewis, B-spline bases for splines under tension, Nu splines, and fractional order splines, Presented at the SIAM-SIGNUM-Meeting, San Francisco, USA, 1975.
- [11] G.M. Nielson, Some piecewise polynomial alternatives to S-splines under tension, *Comput. Aided Geom. Design* (1974) 209–235.
- [12] G.M. Nielson, Rectangular v -splines, *IEEE Comput. Graph. Appl.* 6 (1986) 35–40.
- [13] S. Pruess, Alternatives to the exponential spline in tension, *Math. Comput.* 33 (1979) 1273–1281.
- [14] K. Salkauskas, C^1 splines for interpolation of rapidly varying data, *Rocky Mountain J. Math.* 14 (1984) 239–250.
- [15] M. Sarfraz, A C^2 rational cubic spline alternative to the NURBS, *Comput. Graph.* 16 (1) (1992) 69–78.
- [16] M. Sarfraz, Curves, surfaces for CAD using C^2 rational cubic splines, *Int. J. Eng. Comput.* 11 (2) (1995) 94–102.
- [17] M. Sarfraz, Freeform rational bicubic spline surfaces with tension control, *FACTA UNIVERSITATIS (NIS)*, Series Math. Inform. 9 (1994) 83–93.
- [18] M. Sarfraz, Cubic spline curves with shape control, *Int. J. Comput. Graph.* 18 (5) (1994) 707–713.
- [19] M. Sarfraz, Weighted Nu splines: an alternative to NURBS, advances in geometric modeling, in: M. Sarfraz (Ed.), Wiley, New York, 2003, pp. 81–95, ISBN: 0-470-85937-7.
- [20] I.J. Schoenberg, Contributions to the problem of approximation of equidistant data by analytic functions, *Appl. Math.* 4 (1946) 45–99.
- [21] D.G. Schweiker, An interpolation curve using a spline intension, *J. Math. Phys.* 45 (1996) 312–317.
- [22] M. Sarfraz, Weighted Nu splines with local support basis, *Comput. Graph.* 28 (2004) 539–549.
- [23] M. Sarfraz, Designing objects with a spline, *Int. J. Comput. Math.* 87 (4) (2010) 797–817.
- [24] M.Z. Hussain, M. Hussain, A. Waseem, Shape-preserving trigonometric functions, *Comput. Appl. Math.* 33 (2) (2014) 411–431.
- [25] M. Sarfraz, M.Z. Hussain, M. Hussain, Modeling rational spline for visualization of shaped data, *J. Numer. Math.* 21 (1) (2013) 63–88.
- [26] M.Z. Hussain, M. Hussain, B. Aqeel, Shape-preserving surfaces with constraints on tension parameters, *Appl. Math. Comput.* 247 (2014) 442–464.
- [27] X. Han, Shape preserving piecewise rational interpolant with quartic numerator and quadratic denominator, *Appl. Math. Comput.* 251 (2015) 258–274.